Evaluating the Impact of Historical Lookback Windows on LSTM Predictive Accuracy across the Technology, Healthcare, and Consumer Staples Sectors

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***Abstract*—This study investigates the predictive efficiency of historical stock price patterns on next-day returns, evaluating how accurately Long Short-Term Memory (LSTM) models can forecast adjusted close prices using only historical price data across the Technology, Healthcare, and Consumer Staples sectors. Lookback windows of 10, 30, and 60 trading days were tested to determine whether historical time horizon influences predictive accuracy. This investigation is motivated by the rise of agentic Artificial Intelligence tools that enable retail investors to deploy automated trading strategies using widely accessible historical data. A Many-to-One LSTM architecture was compared against a Linear Regression model and a Naïve Baseline across the specified windows. Results were validated using a one-sided paired t-test within a k=10 TimeSeriesSplit cross-validation framework across three random seeds. The null hypothesis,that the LSTM does not achieve lower MAPE than the Naïve Baseline, could not be rejected across any stock, seed, or window configuration. Both the LSTM and Linear Regression models performed statistically worse than the persistence baseline, providing empirical support for the Efficient Market Hypothesis (EMH) weak form and suggesting that short-term historical price patterns alone are insufficient for next-day stock price prediction.**

I. Introduction

This study investigates the predictive efficiency of historical stock price patterns on next-day returns, specifically evaluating the capacity of Long Short-Term Memory (LSTM) models to forecast adjusted close prices using only historical data. The research focuses on the performance of these models across the Technology, Healthcare, and Consumer Staples industries, utilizing lookback windows of 10, 30, and 60 days. This investigation is timely because the rise of agentic Artificial Intelligence allows retail investors to deploy automated trading strategies that rely on widely accessible historical data. If market predictability is tied to specific historical time horizons, identifying the optimal lookback window becomes a critical factor for feasible wealth management strategies.

To evaluate this research question, a Many-to-One LSTM architecture was compared against both a linear regression model and a naïve baseline across the specified 10, 30, and 60-day windows. The results were validated using a paired t-test within a k=10 TimeSeriesSplit cross-validation framework. The null hypothesis could not be rejected under the one-sided paired t-test framework, confirming that both the LSTM and Linear Regression models performed statistically worse than the naïve baseline when relying solely on historical pricing. Ultimately, these results provide empirical support for the Efficient Market Hypothesis (EMH), suggesting that the "random walk" nature of these assets effectively neutralizes predictive signals derived from past price patterns.

Three companies were selected: Oracle Corporation (ORCL, Technology), Merck & Co. Inc. (MRK, Healthcare), and PepsiCo Inc. (PEP, Consumer Staples), all of which maintain at least 10 years of daily trading data within the dataset. Oracle was selected as a representative of high-growth, high-volatility technology stocks following its transition from a legacy database provider to a major cloud and AI infrastructure company. Johnson & Johnson was initially considered as the Healthcare representative; however, Merck was ultimately selected as a defensive stock whose price behavior is primarily driven by clinical trial outcomes rather than broader news trends. PepsiCo was selected as a Consumer Staples representative, widely regarded as a stable control within the S&P 500 index. The experimental approach involved building and tuning the LSTM model across the three lookback windows before comparing it against the Linear Regression and Naïve Baseline models via k=10-fold cross-validation to determine how accurately each model can predict next-day adjusted close price. The remainder of this paper is organized as follows: Section II details background and related work, Section III details the methodology, Section IV presents the experimental results, and Section V concludes with discussion and future work considerations.

II. Background and Related Work

The pursuit of stock price prediction has evolved from basic statistical forecasting to complex machine learning architectures. However, the core challenge remains the task of separating signal from noise. While early research by Refenes et al. [1] suggested that non-linear neural networks could capture market complexities better than traditional linear methods, the practical application of these models is heavily dictated by the quality of the input data. Shahi et al. [2] and Singh et al. [3] emphasize that while LSTMs are robust, their accuracy often relies on external context or adaptive, real-time tuning. Without these additional layers, a model is forced to find patterns in historical price data, a task that is fundamentally at odds with Fama's [4] Efficient Market Hypothesis (EMH).

The EMH and its associated Random Walk Theory provide the primary theoretical hurdle for this project. Fama [4] argues that in an efficient market, prices fully reflect all available information, meaning future price movements are independent of past trends. This creates a "fair game" environment where any predictable pattern is effectively neutralized by market participants as soon as it is discovered. Timmermann & Granger [5] describe this as a self-destruct mechanism within predictive models. Empirical studies by Htun et al. [6] and Saud & Shakya [7] reinforce this by demonstrating that historical data has a rapidly diminishing predictive shelf life.

This project is built on the architectural logic that "less is more" in a high-volatility environment. Influenced by Guresen et al. [8], who identified that excessive input variables often introduce noise rather than predictive clarity, the feature set was streamlined to focus on core price movements. This study further follows the findings of Saud & Shakya [7] regarding lookback windows; since windows exceeding 10 to 15 days can actively degrade predictive accuracy, this project tests that assumption across shorter and longer windows to determine whether a short-term signal can be isolated. By limiting the model's memory and simplifying its inputs, the goal is to determine whether an LSTM can identify a window of predictability before the market reverts to its random walk state, and whether this varies by industry sector.

III. Methodology

A. Dataset

This study utilized the S&P 500 Stocks dataset sourced from Kaggle [1], which contains daily trading information from January 1, 2010 through December 20, 2024. Three tickers were selected to represent distinct market sectors: Oracle Corporation (ORCL, Technology), Merck & Co. Inc. (MRK, Healthcare), and PepsiCo Inc. (PEP, Consumer Staples). Johnson & Johnson was initially considered as the Healthcare representative; however, it was replaced by Merck after initial dataset inspection revealed insufficient data for the target time window. Oracle was selected as a representative of high-volatility technology stocks, given its history of rapid price movements tied to shifts in the broader technology industry. Merck was selected for its defensive characteristics, as its price behavior is primarily driven by clinical trial outcomes rather than broader market sentiment. PepsiCo was selected as a stable Consumer Staples representative, widely regarded as a consistent performer within the S&P 500 index.

[1] https://www.kaggle.com/datasets/andrewmvd/sp-500-stocks

B. Feature Engineering

The dataset provided the following fields: Date, Symbol, Adjusted Close, Close, High, Low, Open, and Volume. Consistent with the recommendation of Guresen et al. [8] to minimize input complexity and reduce noise, the selected input features were Adjusted Close, Open, High, and Low. Volume was explicitly excluded as it is not a pure price-based field and would likely introduce additional noise. Adjusted Close was selected over raw Close to account for corporate actions such as stock splits, dividends, and new share issuances, which are critical to accurately representing a company's historical price trajectory. A sliding window function was implemented to transform the raw time series into a supervised learning format, creating sequences of price observations that the model can learn from. Lookback windows of 10, 30, and 60 trading days were tested to evaluate the effect of historical horizon on predictive accuracy.

C. Data Preparation

A chronological 70%/15%/15% train, validation, and test split was applied across the full 10-year historical dataset. The chronological ordering is critical when working with sliding lookback windows, as random shuffling would introduce look-ahead bias and invalidate the temporal integrity of the evaluation. Data normalization was performed using the `MinMaxScaler` from the Scikit-Learn package, scaling all features to a range suitable for the LSTM's internal activation functions. To prevent data leakage, the scaler was fitted exclusively on the training partition; the resulting parameters were then applied to transform the validation and test sets, ensuring no future information was present during model training.

D. Model Architecture

Three models were implemented to provide a comprehensive evaluation framework:

Naive Baseline: The persistence model predicts tomorrow's adjusted close price as equal to today's adjusted close price. This model serves as the floor benchmark, directly embodying the Random Walk implied by Fama's [4] Efficient Market Hypothesis. All proposed models are evaluated against this baseline.

Linear Regression: Implemented via Scikit-Learn, the linear regression model accepts a flattened sliding window as its input vector, treating each historical day as an independent variable. This model serves as a traditional machine learning benchmark to determine whether the LSTM's sequential architecture provides additional predictive value beyond a simpler linear approach.

Long Short-Term Memory (LSTM): The LSTM model was implemented using TensorFlow Keras with a Many-to-One architecture, ingesting a sequence of historical price observations and producing a single next-day adjusted close prediction. The architecture consists of three layers: an LSTM layer, a Dropout layer, and a Dense output layer. The LSTM layer processes the sliding window sequences to identify temporal patterns in historical price data. A Dropout rate of 0.2 was applied to mitigate the effects of high market volatility and noise, consistent with the findings of Guresen et al. [8]. The Dense layer performs a weighted sum to condense the LSTM output into a single predicted price value. Weight initialization used `glorot_uniform` for kernel weights and `orthogonal` for recurrent weights to provide a stable starting point for longer sequence training. Gradient clipping was applied at `clipnorm=1.0` to prevent exploding gradients. The Adam optimizer was configured with a learning rate of 0.0001, and Early Stopping with a patience of 50 epochs was used to prevent premature convergence.

E. Evaluation Framework

Three metrics were used to evaluate model performance. Mean Absolute Error (MAE) provides an interpretable dollar-value measure of average prediction error. Root Mean Squared Error (RMSE) serves as a penalty metric that amplifies the impact of large prediction errors, providing additional context for outlier sensitivity. Mean Absolute Percentage Error (MAPE) was designated as the primary evaluation metric, as it expresses error as a percentage of the actual price, enabling scale-independent comparison across stocks trading at different price levels.

It is important to distinguish between the loss function used during training and the performance measures used for evaluation. Mean Squared Error (MSE) serves as the training loss function, selected for its compatibility with gradient-based optimization within the LSTM architecture. MSE acts as the internal guide the model uses to measure how well it is learning during each epoch, with a declining MSE curve confirming that the model is actively updating its weights rather than stagnating. MAPE, MAE, and RMSE are provided strictly as post-training evaluation metrics, with MAPE designated as the primary measure given its scale independence and ability to communicate error in plain percentage terms across stocks trading at very different price levels.

To ensure statistical rigor, a k=10 TimeSeriesSplit cross-validation was employed in place of standard randomized k-fold. This approach maintains strict chronological ordering across folds, simulating a real-world walk-forward forecasting environment and preventing look-ahead bias. Three random seeds (42, 99, 123) were used to account for stochastic variation in LSTM weight initialization and to confirm result reproducibility across different initialization conditions.

F. Statistical Hypothesis

The primary hypothesis of this study was that a tuned Many-to-One LSTM model, using a reduced historical price feature set and lookback windows of 10, 30, or 60 trading days, would outperform a naive persistence baseline in predicting next-day adjusted closing prices. This expectation was based on the assumption that the LSTM architecture could identify short-term temporal patterns in historical price behavior that would not be captured by a simple persistence model. A secondary comparison against Linear Regression was included as a traditional machine learning benchmark to determine whether the LSTM's sequential architecture provided additional predictive value beyond a simpler linear approach.

The null hypothesis states: the LSTM model does not achieve statistically significantly lower MAPE than the Naive Baseline, that is, $H_0 : \mu_{LSTM} \geq \mu_{Naive}$. The alternative hypothesis states: the LSTM model achieves statistically significantly lower MAPE than the Naive Baseline, that is, $H_1 : \mu_{LSTM} < \mu_{Naive}$. A one-sided paired Student's t-test was selected via `scipy.stats.ttest_rel`, as the hypothesis was directional, the study anticipated that the LSTM would achieve lower error than the persistence baseline. A standard alpha of 0.05 was applied. Given six total comparisons across three stocks and two model pairs, a Bonferroni correction was applied, yielding an adjusted significance threshold of $\alpha = 0.0083$.

IV. Experiments and Results

A. Single Holdout Evaluation

The single holdout evaluation served as an initial blind test to compare the Naive Baseline, Linear Regression, and LSTM models across 10-, 30-, and 60-day lookback windows prior to the more rigorous k=10 TimeSeriesSplit evaluation. Results are summarized in Table I. The Naive Baseline and Linear Regression models remained stable across all three windows, while the LSTM produced evaluable results only for the 10-day window, experiencing structural convergence failure for the 30- and 60-day windows.

The Naive Baseline produced the lowest MAPE values across all three stocks, averaging 1.27% for ORCL, 0.91% for MRK, and 0.75% for PEP on the 10-day window. Linear Regression slightly exceeded the Naive Baseline, averaging 1.32% for ORCL, 0.97% for MRK, and 0.83% for PEP. The LSTM produced substantially higher error on the 10-day window, averaging 16.70% MAPE for ORCL, 17.29% for MRK, and 10.15% for PEP. These elevated single holdout LSTM values reflect training from a single seed without the extended hyperparameter tuning applied in the full experimental notebook. The 30- and 60-day LSTM windows experienced structural convergence failure, consistent with the findings of Saud & Shakya [7] that lookback windows exceeding 10 to 15 days degrade predictive accuracy. This finding aligns with the project's design assumption that longer lookback windows may introduce more historical noise than useful predictive signal.

B. K-Fold Cross-Validation Results

The k=10 TimeSeriesSplit evaluation tested each model under a validation structure that preserved chronological order

TABLE I
SINGLE HOLDOUT EVALUATION — MAPE (%) BY STOCK AND WINDOW

Stock	Model	W10	W30	W60
ORCL	Naive Baseline	1.27	1.25	1.25
	Linear Regression	1.32	1.32	1.40
	LSTM	16.70	N/A	N/A
MRK	Naive Baseline	0.91	0.90	0.89
	Linear Regression	0.97	0.97	1.01
	LSTM	17.29	N/A	N/A
PEP	Naive Baseline	0.75	0.73	0.73
	Linear Regression	0.83	0.87	0.90
	LSTM	10.15	N/A	N/A

and reduced the risk of look-ahead bias. Each model was evaluated across 10 sequential folds and three random seeds (42, 99, 123), providing a robust assessment of performance stability across different training and evaluation windows.

The Naive Baseline produced the strongest and most stable results across all three stocks, averaging 1.12% MAPE for ORCL, 0.92% for MRK, and 0.74% for PEP across all seeds. As a deterministic model, the Naive Baseline exhibited zero cross-seed variance, confirming its stability as a persistence benchmark.

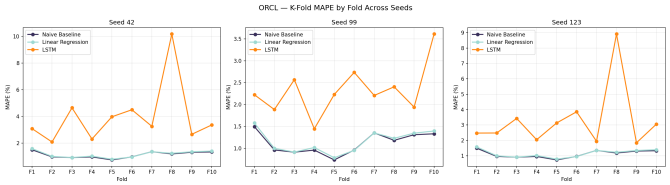

Fig. 1. K-Fold MAPE by Fold Across Seeds — ORCL (Window 10)

Linear Regression performed marginally worse than the Naive Baseline across all stocks and seeds, averaging 1.16% MAPE for ORCL, 0.98% for MRK, and 0.77% for PEP. The flattened lookback window provided insufficient additional predictive signal to outperform the simple persistence assumption, as illustrated in Figs. 1, 2, and 4.

To verify that the LSTM architecture was actively learning rather than failing to train, the loss history for a representative fold was captured and examined. Figure 5 presents the training and validation MSE loss curves for ORCL, Seed 99, Fold 10. Training loss declined consistently across 72 epochs before early stopping restored best weights at the epoch of minimum validation loss. The validation loss plateaued early and remained stable, confirming that the model trained successfully but simply ran out of useful signal from the available price-based data, consistent with Random Walk properties rather than a failure of the training process.

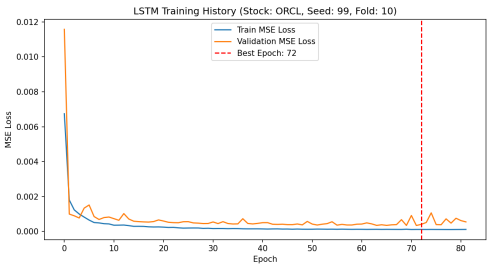

Fig. 3. LSTM Training Curve — ORCL, Seed 99, Fold 10. Training MSE loss (blue) declined across 72 epochs. Validation MSE loss (orange) plateaued early, with early stopping restoring best weights at epoch 72.

The LSTM model exhibited substantially higher error and greater cross-seed instability than both comparison models, averaging 3.21% MAPE for ORCL, 3.25% for MRK, and 3.19% for PEP, with seed-level standard deviations of 0.69, 0.30, and 0.95, respectively. The elevated and variable MAPE across folds and seeds indicates that the LSTM architecture was more sensitive to weight initialization than either the Naive Baseline or Linear Regression under the price-only feature constraint.

C. Cross-Seed Stability

The Naive Baseline and Linear Regression models produced identical results across all three seeds, with a cross-seed standard deviation of 0.00% for all stocks, confirming that these deterministic models are unaffected by random initialization. The LSTM exhibited meaningful cross-seed variance, with standard deviations of 0.69%, 0.30%, and 0.95% for ORCL, MRK, and PEP, respectively. Seed 42 consistently produced the highest mean MAPE and standard deviation across all three stocks, suggesting that this initialization was particularly unfavorable for the LSTM architecture under a price-only feature constraint. This variability underscores the importance of multi-seed evaluation for deep learning models, as a single seed may not be representative of the architecture's true performance distribution. As shown in Fig. 5, the LSTM's mean MAPE consistently exceeded both comparison models across all seeds and stocks.

D. Statistical Significance

All p-values exceeded the Bonferroni-corrected significance threshold of $\alpha = 0.0083$, confirming that the null hypothesis $H_0 : \mu_{LSTM} \geq \mu_{Naive}$ could not be rejected across any stock or seed configuration. Neither the LSTM nor the Linear Regression model achieved statistically significantly lower MAPE than the Naive Baseline under the one-sided paired t-test framework. The consistently high p-values (ranging from 0.9657 to 0.9999) indicate that both proposed models performed in the opposite direction of the stated hypothesis — statistically significantly worse than the persistence baseline rather than better. These results provide empirical support for Fama's [4] Efficient Market Hypothesis weak form, suggesting that short-term historical price patterns do not contain sufficient signal to outperform a simple persistence model when

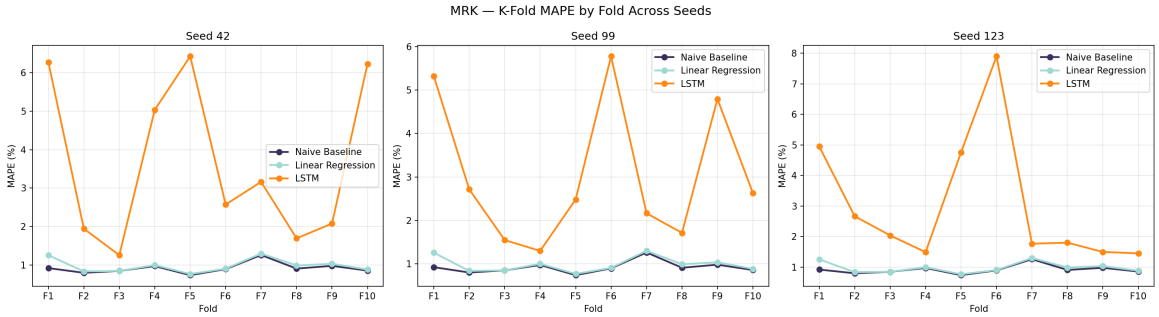

Fig. 2. K-Fold MAPE by Fold Across Seeds — MRK (Window 10)

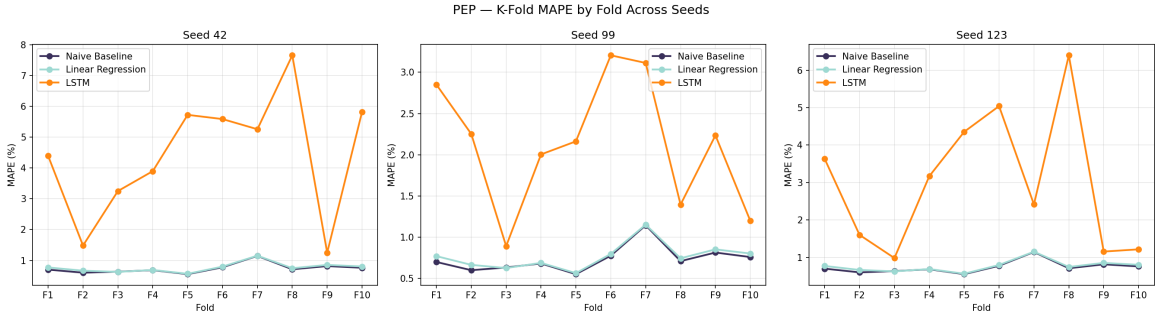

Fig. 4. K-Fold MAPE by Fold Across Seeds — PEP (Window 10)

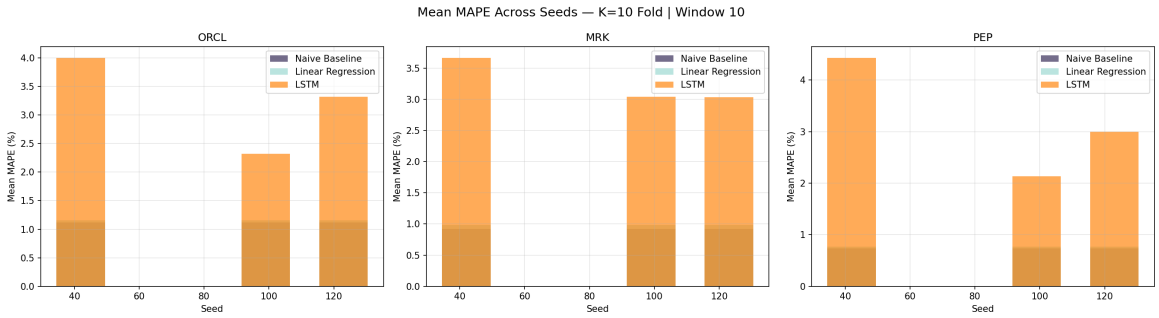

Fig. 5. Mean MAPE Across Seeds — K=10 Fold, Window 10

using price-only features across Technology, Healthcare, and Consumer Staples sectors.

V. Discussion and Conclusion

A. Discussion

The results of this study consistently support the Efficient Market Hypothesis weak form as formalized by Fama [4]. The Naive Baseline outperformed both the Linear Regression and LSTM models across all three sectors and all evaluated configurations, confirming that yesterday's adjusted close price already accounts for the information available to predict today's price. The Random Walk nature of these assets effectively neutralized any predictive signal the LSTM attempted to extract from historical price patterns. It is strongly worth noting that a result in the opposite direction, where the LSTM model was able to significantly outperform the naive baseline, would have warranted heavy scrutiny rather than celebration, based on the volume and weight of Efficient Market Hypothesis literature and the price-only feature constraint, as such a result would

have indicated data leakage or look-ahead bias rather than true predictive power. The fact that the results aligned with theoretical expectations served as a credibility check on the evaluation framework itself.

Furthermore, the degradation observed in longer lookback windows suggests that additional historical context introduced more noise than useful signal, consistent with the findings of Saud & Shakya [7] that windows exceeding 10 to 15 days actively degrade predictive accuracy.

Additional research on comparable LSTM implementations has reported consistent findings, where naive persistence baselines matched or outperformed LSTM models on price-only stock prediction tasks. While more complex architectures have closed the gap to within a competitive range, the core finding remains consistent across studies as LSTM models relying solely on historical price data are unable to outperform the naive baseline. This is a result that aligns with EMH weak form and suggests this ceiling is a property of the input data rather than a limitation of any specific architecture.

One unexpected finding emerged from the cross-sector comparison. Despite Oracle Corporation being selected as the highest-volatility stock in the study, the LSTM model produced its lowest mean MAPE for ORCL compared to MRK and PEP across several seed configurations. This was counter-intuitive given the assumption that high volatility would make price prediction more difficult. One possible explanation is that Oracle's strong directional trend during the test period (2022-2024), driven by its transition to cloud and AI infrastructure, provided a more learnable momentum signal than the mean-reverting behavior exhibited by PEP during the same period. This suggests that sector volatility alone may not be the primary determinant of LSTM predictability, market regime and directionality may play a more significant role.

B. Limitations

This study was subject to two primary limitations. First, the feature set was intentionally restricted to price-only variables, excluding external signals such as trading volume, sentiment data, and macroeconomic indicators. While this constraint was intentional, designed to isolate the predictive value of historical price patterns, it also meant the models lacked the contextual information needed to anticipate significant price movements driven by earnings announcements, clinical trial results, or broader market events. Second, the dataset was limited to daily trading data. Intraday data at hourly or sub-hourly resolution may contain stronger short-term autocorrelation patterns that are better suited to sequence-based prediction, as the signal-to-noise ratio at finer time intervals may be more favorable for LSTM-style architectures.

C. Future Work

Future work would focus on two primary directions. First, incorporating non-price features such as financial news sentiment scores via tools like FinBERT or VADER, trading volume, and technical indicators such as RSI and MACD would provide the model with contextual signals beyond raw price behavior. These features may capture the information that drives deviations from the random walk, specific events that price-only models cannot anticipate. Second, alternative recurrent architectures such as the Gated Recurrent Unit (GRU) would be worth evaluating, as GRU has fewer trainable parameters than LSTM and has been shown by Saud & Shakya [7] to outperform LSTM on similar time series tasks. A hybrid CNN-LSTM or attention-based architecture could also be explored to improve the model's ability to weight recent price behavior more heavily than distant observations.

D. Conclusion

This study evaluated whether a single-layer Many-to-One LSTM model could predict next-day adjusted close prices using only historical price-based features across Technology, Healthcare, and Consumer Staples sectors. Three lookback windows (10, 30, and 60 days) were tested against a Naive Baseline and Linear Regression benchmark using k=10 TimeSeriesSplit cross-validation across three random seeds. A one-sided paired Student's t-test with Bonferroni correction confirmed that neither the LSTM nor the Linear Regression model achieved statistically significantly lower MAPE than the persistence baseline. The null hypothesis could not be rejected across any stock, seed, or window configuration. These findings are consistent with the Efficient Market Hypothesis weak form and empirically demonstrate that short-term historical price patterns alone are insufficient for next-day stock price prediction. Future work should incorporate sentiment data, technical indicators, and alternative recurrent architectures to determine whether richer feature sets can overcome the predictive ceiling imposed by market efficiency.

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Appendix

Claude.ai (Anthropic) and ChatGPT (OpenAI) were used as development assistants throughout this project. These tools were used for code debugging, architectural decision review, conceptual explanation of machine learning topics, LaTeX

formatting guidance, and report structure feedback. All code was written, reviewed, and modified by the author. All analytical decisions, interpretations, and conclusions are the author's own.

AI assistance was not used to generate results, fabricate data, or produce final written content directly. Drafted sections were reviewed, rewritten, and approved by the author prior to inclusion. The use of these tools is consistent with the course GenAI Kit disclosure requirements.